EE 533 Information Theory

Week 02

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Definition (Markov Chain)

Random variables are said to form a Markov Chain (denoted by $\mathrm{X} \to \mathrm{Y} \to \mathrm{Z}$) iff the conditional distribution of Z given X and Y is independent of X, i.e.

$$p_{\mathrm{Z}|\mathrm{Y},\mathrm{X}}(z|y,x) = p_{\mathrm{Z}|\mathrm{Y}}(z|y)\,.$$

Since $p_{\mathrm{X},\mathrm{Z}|\mathrm{Y}}(x,z|y) = p_{\mathrm{X}|\mathrm{Y}}(x|y)\,p_{\mathrm{Z}|\mathrm{Y},\mathrm{X}}(z|y,x)$ this is equivalent to:

$$p_{\mathrm{X},\mathrm{Z}|\mathrm{Y}}(x,z|y) = p_{\mathrm{X}|\mathrm{Y}}(x|y)\,p_{\mathrm{Z}|\mathrm{Y}}(z|y)$$

Theorem

(a) $\mathrm{X} \to \mathrm{Y} \to \mathrm{Z}$ *iff* X *and* Z *are conditionally independent given* Y.

(b) $\mathrm{X} \to \mathrm{Y} \to \mathrm{Z}$ *iff* $I\,(\mathrm{X};\mathrm{Z}|\mathrm{Y}) = 0$.

(c) $\mathrm{X} \to \mathrm{Y} \to \mathrm{Z}$ *iff* $\mathrm{Z} \to \mathrm{Y} \to \mathrm{X}$.

Because of part (c) Markov chains are also denoted by $\mathrm{X} \leftrightarrow \mathrm{Y} \leftrightarrow \mathrm{Z}$

Example

If $Z = f(Y)$, then $X \to Y \to Z$ because $H(f(Y)|Y) = 0$ and thus $I(X; f(Y)|Y) = 0$.

Example

If X, Y, and Z are binary random variables such that

$$p_X(0) = 1/2 \qquad p_X(1) = 1/2$$
$$p_{Y|X}(x|x) = 3/5 \qquad p_{Y|X}(1-x|x) = 2/5$$
$$p_{Z|Y,X}(y|x,y) = 2/3 \qquad p_{Z|Y,X}(1-y|x,y) = 1/3$$

then $X \to Y \to Z$.

Theorem

If $X \to Y \to Z$, *then*

a) $I(X;Y) \geq I(X;Z)$ *(the Data Processing Inequality)*

b) $I(X;Y) \geq I(X;Y|Z)$.

Proof.

$$
\begin{aligned}
I(X;Y) &= I(X;Y) + I(X;Z|Y) && \text{because } X \to Y \to Z \text{ implies } I(X;Z|Y) = 0 \\
&= I(X;Y,Z) && \text{by the chain rule for the mutual information} \\
&= I(X;Z) + I(X;Y|Z) && \text{by the chain rule for the mutual information} \\
&\geq I(X;Z) \vee I(X;Y|Z) && \text{by the non-negativity of M.I. and C.M.I}
\end{aligned}
$$

□

HW: Find binary R.V.'s X, Y, Z s.t. $I(X;Y) \leq I(X;Y|Z)$.

Corollary

If $X \to Y \to Z$, *then* $H(X|Y) \leq H(X|Z)$

Corollary

$I(X;Y) \geq I(X;f(Y))$

Definition (Sufficient Statistic)

Let $\{p_{\mathsf{X}}^{\theta}\}$ be a family probability distributions for X indexed by θ, the conditional probability distribution of T given X and Θ be $p_{\mathsf{T}|\mathsf{X}}$, i.e. $p_{\mathsf{T}|\mathsf{X}}^{\theta} = p_{\mathsf{T}|\mathsf{X}}$ (equivalently $\Theta \to \mathsf{X} \to \mathsf{T}$ for all p_{Θ}), and $p_{\mathsf{T}}^{\theta}(t) := \sum_x p_{\mathsf{T}|\mathsf{X}}(t|x)\, p_{\mathsf{X}}^{\theta}(x)$. Then T is a sufficient statistic (s.s.) of X for θ iff there exists a $p_{\mathsf{X}|\mathsf{T}}$ such that $p_{\mathsf{X}}^{\theta} p_{\mathsf{T}|\mathsf{X}} = p_{\mathsf{T}}^{\theta} p_{\mathsf{X}|\mathsf{T}}$, i.e. such that $p_{\mathsf{X}|\mathsf{T}}^{\theta} = p_{\mathsf{X}|\mathsf{T}}$.

Theorem

If $\Theta \to \mathsf{X} \to \mathsf{T}$ for all p_{Θ}, then the following are equivalent:

i) T *is a sufficient statistic of* X *for* Θ.

ii) $\Theta \to \mathsf{T} \to \mathsf{X}$ *for all* p_{Θ}.

iii) $I(\Theta; \mathsf{X}|\mathsf{T}) = 0$ *for all* p_{Θ}.

iv) $I(\Theta; \mathsf{X}) = I(\Theta; \mathsf{T})$ *for all* p_{Θ}.

Proof of "T is a s.s. of X for Θ" $\Rightarrow$ "$I(\Theta; X|T) = 0$ for all p_Θ".

T is a s.s. of X for Θ then $p^\theta_{X|T} = p_{X|T}$, i.e., $p_{X|T,\Theta} = p_{X|T}$ for all p_Θ, by definition.

Hence $\Theta \to T \to X$ for all p_Θ and thus $I(\Theta; X|T) = 0$ for all p_Θ. $\square$

Proof of "$I(\Theta; X|T) = 0$ for all p_Θ" $\Rightarrow$ "T is a s.s. of X for Θ".

$I(\Theta; X|T = t) = 0$ for a p_Θ iff Θ and X are conditionally independent given $T = t$.

$I(\Theta; X|T = t) = 0$ for a p_Θ iff $p_{X|T}(\cdot|t) = p_{X|T,\Theta}(\cdot|t, \theta)$ for all (θ, t) s.t. $p_\Theta(\theta)\, p^\theta_T(t) > 0$.

$$I(\Theta; X|T) = \sum_t p_T(t)\, I(\Theta; X|T = t)$$

Then $I(\Theta; X|T) = 0$ for all p_Θ iff $p_{X|T}(\cdot|t) = p_{X|T,\Theta}(\cdot|t, \theta)$ for all t s.t. $p^\theta_T(t) > 0$ for some θ.

$\square$

Example ([CT06, p.36])

- Let the unknown parameter θ be the probability of a head for a possibility biased coin.
- Let $X_1, X_2, \ldots, X_n$ be binary random variables associated with the n independent tosses of this coin such that $X_j = 1$ iff j^{th} toss is a head
- Let T be the number of heads in these n tosses, i.e $T := \sum_{j=1}^{n} X_j$

Then

$$p^{\theta}_{X_1^n}(x_1^n) = \prod_{j=1}^{n} \theta^{x_j}(1-\theta)^{1-x_j} \qquad = \theta^t(1-\theta)^{n-t}$$

$$p_{T|X_1^n}(t|x_1^n) = \mathbb{1}_{\{t=\sum_{i=1}^{n} x_i\}}$$

$$p^{\theta}_{T}(t) := \sum_{x_1^n} p_{T|X_1^n}(t|x_1^n)\, p^{\theta}_{X_1^n}(x_1^n) \qquad = \binom{n}{t}\theta^t(1-\theta)^{n-t}$$

$$p_{X_1^n|T,\Theta}(x_1^n, |t, \theta) = \frac{p^{\theta}_{X_1^n}(x_1^n)\, p_{T|X_1^n}(t|x_1^n)}{p^{\theta}_{T}(t)} \qquad = \frac{1}{\binom{n}{t}}\mathbb{1}_{\{\sum_{i=1}^{n} x_i = t\}}.$$

Thus $p_{X_1^n|T,\Theta} = p_{X_1^n|T}$, i.e. T is a sufficient statistic of X for θ.

Fano's Inequality

X: a discrete r.v. to be guessed(decoded) that takes values in a finite set $\mathcal{X}$.

Y: observation of about X that is used to guess/decode.

Z: $\mathcal{X}$ value guess based on Y.

Theorem

$$H(\mathsf{X}|\mathsf{Y}) \leq \hbar(P_{\mathbf{e}}) + P_{\mathbf{e}} \log(|\mathcal{X}| - 1) \qquad \textit{where } P_{\mathbf{e}} = \mathbf{P}(\mathsf{Z} \neq \mathsf{X}).$$

Proof.

$\mathsf{X} \to \mathsf{Y} \to \mathsf{Z}$ by construction. Thus $I(\mathsf{X};\mathsf{Y}) \geq I(\mathsf{X};\mathsf{Z})$ and $H(\mathsf{X}|\mathsf{Y}) \leq H(\mathsf{X}|\mathsf{Z})$ by DPI.
Let $\mathsf{E} = \mathbb{1}_{\{\mathsf{Z} \neq \mathsf{X}\}}$. Then

$$\begin{aligned}
H(\mathsf{X}|\mathsf{Z}) &= H(\mathsf{E},\mathsf{X}|\mathsf{Z}) && \text{by the chain rule and } H(\mathsf{E}|\mathsf{X},\mathsf{Z}) = 0 \\
&= H(\mathsf{E}|\mathsf{Z}) + H(\mathsf{X}|\mathsf{Z},\mathsf{E}) && \text{by the chain rule for the entropy} \\
&\leq H(\mathsf{E}) + H(\mathsf{X}|\mathsf{Z},\mathsf{E}) && \text{conditioning decreases the entropy} \\
&= \hbar(P_{\mathbf{e}}) + P_{\mathbf{e}} H(\mathsf{X}|\mathsf{Z},\mathsf{E}=1) && \text{because } H(\mathsf{X}|\mathsf{Z},\mathsf{E}=0) = 0 \\
&\leq \hbar(P_{\mathbf{e}}) + P_{\mathbf{e}} \log(|\mathcal{X}| - 1) && \text{why?} \mathsf{E} = 1 \Rightarrow \mathsf{X} \neq \mathsf{Z}.
\end{aligned}$$

□

For functions on p.m.f.'s let us adopt the notation $G(p_X) = G(p_X(1), p_X(2), \ldots, p_X(k))$. Then

(i) $G(p_1, \ldots, p_n)$ is a symmetric function of its variables, i.e. invariant under permutations, for $n = 2, 3, \ldots$. (recall the definition)

(ii) $G(p, 1-p)$ is continuous in p. (recall $\hbar(p)$)

(iii) $G(1/2, 1/2) = 1$. (recall $\hbar(p)$)

(iv) $G(tp_1, (1-t)p_1, p_2 \ldots, p_n) = G(p_1, \ldots, p_n) + p_1 G(t, (1-t))$. (def.)

(v) $G(p_1, \ldots, p_n) + G(q_1, \ldots, q_k) = G(p_1 q_1, \ldots, p_1 q_k, p_2 q_1, \ldots, p_n q_k)$.
($H(X, Y) = H(X) + H(Y)$ for independent X and Y)

Theorem ([Fad56])
$H(p_X)$ is the only function that satisfies the axioms (i),(ii),(iii),(iv).

There are other axiomatic characterizations of $H(p_X)$, see [AD75], [CK11, p. 12-13].

Theorem ([Rén61])
Axioms (i),(ii),(iii),(v) *are satisfied only by $H(p_X)$ and functions of the form*

$$H_\alpha(p_X) := \frac{1}{1-\alpha} \log \left(\sum_x (p_X(x))^\alpha \right)$$

for some $\alpha > 0$ other than one, called the order α Rényi entropy.

Neighborhood and Convergence

A sequence of real number $x_1, x_2, \ldots$ converges to x, i.e. $x_n \to x$ as $n \to \infty$, iff $\forall \epsilon > 0\ \exists n \in \mathbb{Z}_+$ s.t. $|x_k - x| < \epsilon$ for all $k \geq n$.

In plain English: For every neighborhood $\mathcal{N}$ of the point x there exists an n such that all x_k's are the neighborhood $\mathcal{N}$ for $k \geq n$.

When points in considerations are more sophisticated mathematical objects, there are different ways to define the neighborhoods and hence the convergence.

Example

Let $f_1, f_2, \ldots$ be a sequence of integrable functions on $(0, 1)$.

(i) *Pointwise convergence:* $f_n(x) \to f(x)$ as $n \to \infty$ for all $x \in (0, 1)$

(ii) *Convergence in norm:* $\int_0^1 |f_n(x) - f(x)| \mathrm{d}x \to 0$ as $n \to \infty$.

(iii) *Uniform Convergence:* $\forall \epsilon > 0 \exists n \in \mathbb{Z}_+ s.t. \sup_{x \in [0,1]} |f_k(x) - f(x)| \leq \epsilon\ \forall k \geq n$.

- $f_n(x) := n\mathbb{1}_{\{x \in (0, 1/n)\}} \Rightarrow f_n$ converges to 0 pointwise but not in norm.
- $g_n(x) := \mathbb{1}_{\{x \in (t - 1/2n, t + 1/2n)\}} \Rightarrow g_n$ converges to 0 in norm but not pointwise.
- Uniform convergence $\Rightarrow$pointwise convergence and convergence in norm.

X is a r.v. in $(\Omega, \mathcal{F}, \mathbf{P})$ iff $X : \Omega \to \mathbb{R}$ s.t. $X^{-1}((a,b)) \in \mathcal{F}$ for all $a, b \in \mathbb{R}$.

Definition ([Shi95, Ch.2§10])

Let $X_1, X_2, \ldots,$ and X be r.v.s in the probability space $(\Omega, \mathcal{F}, \mathbf{P})$.

- $X_n \to X$ almost surely, i.e. $X_n \xrightarrow{a.s.} X$, iff $\mathcal{C} := \{\omega \in \Omega : X_n(\omega) \to X(\omega)\}$ is an event whose probability is 1, i.e. $\mathcal{C} \in \mathcal{F}$ and $\mathbf{P}(\mathcal{C}) = 1$.
  $X_n \xrightarrow{a.s.} X$ is also denoted by called $X_n \xrightarrow{\mathbf{P}-\mathbf{a.s.}} X$.
- $X_n \to X$ in rth mean where $0 < r < \infty$, i.e. $X_n \xrightarrow{r} X$, iff $\mathbf{E}\left[|X_n^r|\right] < \infty$ for all n and $\mathbf{E}\left[|X_n - X|^r\right] \to 0$ as $n \to \infty$
  $X_n \xrightarrow{1} X$ is also called $X_n \to X$ in mean, or l.i.m.$X_n = X$.
  $X_n \xrightarrow{2} X$ is also called in mean square, or $X_n \xrightarrow{m.s.} X$.
- $X_n \to X$ in probability, i.e. $X_n \xrightarrow{\mathbf{P}} X$, iff $\mathbf{P}(|X_n - X| > \epsilon) \to 0$ as $n \to \infty$ for all $\epsilon > 0$.
- $X_n \to X$ in distribution iff $\mathbf{P}(X_n \le x) \to \mathbf{P}(X \le x)$ as $n \to \infty$ for all points x at which the function $F_X(x) := \mathbf{P}(X \le x)$ is continuous.

Theorem ([Shi95, Ch.2§10])

a) $(X_n \xrightarrow{a.s.} X) \Rightarrow (X_n \xrightarrow{\mathbf{P}} X)$. *advanced [GS+01, p. 312]*

b) $(X_n \xrightarrow{r} X\ ,\ r > 0) \Rightarrow (X_n \xrightarrow{\mathbf{P}} X)$. *Markov's inequality.*

c) $(X_n \xrightarrow{\mathbf{P}} X) \Rightarrow X_n$ *converges to* X *in distribution.* *def.'s [GS+01, p. 311]*

d) *If* $s > r > 0$ *then* $(X_n \xrightarrow{s} X) \Rightarrow (X_n \xrightarrow{r} X)$. *Jensen's inequality.*

e) *Converse implications do not hold in general.*

Theorem ([Shi95, Ch.2§10])

a) *If* $X_n \xrightarrow{D} c$, *where* c *is a constant then* $X_n \xrightarrow{\mathbf{P}} c$.

b) *If* $X_n \xrightarrow{\mathbf{P}} X$ *and* $\exists k$ *s.t.* $\mathbf{P}(|X_n| \leq k) = 1\ \forall n$, *then* $X_n \xrightarrow{r} X\ \forall r \geq 1$.

c) *If* $\sum_n \mathbf{P}(|X_n - X| > \epsilon) < \infty$ *for all* ϵ, *then* $X_n \xrightarrow{a.s} X$.

d) *If* $X_n \geq 0$ *for all* n, $X_n \xrightarrow{a.s.} X < \infty$, *and* $\mathbf{E}[X_n] \to \mathbf{E}[X] < \infty$ *then* $X_n \xrightarrow{1} X$.

Theorem ((Weak) Law of Large Numbers, [Shi95, p.325])

Let $X_1, X_2, \ldots$ *be independent identically distributed random variables with* $\mathbf{E}[|X_1|] < \infty$, $S_n := X_1 + \ldots + X_n$, *and* $\mu := \mathbf{E}[X_1]$, *then* $\frac{S_n}{n} \xrightarrow{\mathbf{P}} \mu$, *that is, for every* $\epsilon > 0$

$$\mathbf{P}\left(\left|\frac{S_n}{n} - \mu\right| > \epsilon\right) \to 0 \qquad \text{as } n \to \infty.$$

Theorem (Strong Law of Large Numbers, [Shi95, p.391])

Let $X_1, X_2, \ldots$ *be independent identically distributed random variables with* $\mathbf{E}[|X_1|] < \infty$, $S_n := X_1 + \ldots + X_n$, *and* $\mu := \mathbf{E}[X_1]$, *then* $\frac{S_n}{n} \xrightarrow{a.s.} \mu$, *that is*

$$\mathbf{P}\left(\left\{\omega \in \Omega : \frac{S_n(\omega)}{n} \to \mu \text{ as } n \to \infty\right\}\right) = 1.$$

Markov's inequality: If $X \geq 0$, then

$$\mathbf{P}(X \geq \gamma) \leq \frac{\mathbf{E}[X]}{\gamma} \qquad \forall \gamma > 0. \tag{1}$$

Proof.

$$\begin{aligned}\mathbf{E}[X] &= \mathbf{E}[X|X \geq \gamma]\,\mathbf{P}(X \geq \gamma) + \mathbf{E}[X|X < \gamma]\,\mathbf{P}(X < \gamma) \\ &\geq \mathbf{E}[X|X \geq \gamma]\,\mathbf{P}(X \geq \gamma) \\ &\geq \gamma \mathbf{P}(X \geq \gamma)\,.\end{aligned}$$

□

Chebyshev's inequality: If $\mathbf{E}[X] = \mu$ and $\mathbf{E}\left[(X-\mu)^2\right] = \sigma^2$, then

$$\mathbf{P}(|X-\mu| \geq \gamma) \leq \frac{\sigma^2}{\gamma^2} \qquad \forall \gamma > 0. \tag{2}$$

Proof.

Apply Markov's inequality to $Y = (X - \mathbf{E}[X])^2$. □

Chernoff bound:

$$\mathbf{P}(X \geq \gamma) \leq \frac{\mathbf{E}\left[e^{\tau X}\right]}{e^{\tau\gamma}} \qquad \forall \tau > 0. \tag{3}$$

Proof.

Apply Markov's inequality to $Y = e^{\tau X}$. □

Why $\tau \geq 0$? What happens when $\tau \leq 0$?

Proof of Weak Law of Large Numbers with $\mathbf{E}\left[(X_1)^2\right] < \infty$ hypothesis.

$$\begin{aligned}
\mu &:= \mathbf{E}\left[X_1\right] \\
\sigma^2 &:= \mathbf{E}\left[(X_1 - \mu)^2\right] \\
\mathbf{E}\left[S_n\right] &= n\mu \\
\mathbf{E}\left[(S_n - n\mu)^2\right] &= \mathbf{E}\left[\left[\sum\nolimits_{k=1}^{n}(X_k - \mu)\right]^2\right] \\
&= \sum\nolimits_{k,j} \mathbf{E}\left[(X_k - \mu)(X_j - \mu)\right] \\
&= n\sigma^2 \\
\mathbf{P}(|S_n - n\mu| \geq \gamma) &\leq \tfrac{n\sigma^2}{\gamma^2} && \text{by Chebysev's inequality} \\
\mathbf{P}\left(\left|\tfrac{S_n}{n} - \mu\right| \geq \epsilon\right) &\leq \tfrac{\sigma^2}{\epsilon n} && \text{for } \gamma = n\epsilon
\end{aligned}$$

□

Corollary

Let $X_1, X_2, \ldots$ be independent zero mean random variables with finite second moments satisfying $\left(\frac{1}{n^2}\sum_{k=1}^{n}\sigma_k^2\right) \to 0$ as $n \to \infty$ where $\sigma_k^2 := \mathbf{E}\left[(X_k)^2\right]$, then $\frac{S_n}{n} \xrightarrow{\mathbf{P}} 0$.

In fact $\sum_n \frac{\sigma_n^2}{n^2} < \infty$ implies $\frac{S_n}{n} \xrightarrow{\mathbf{P}-\text{a.s.}} 0$ by [Shi95, p.389].

Theorem (Central Limit Theorem [Shi95, p.326])

Let $X_1, X_2, \ldots$ *be independent identically distributed nondegenerate (i.e.* $\sigma > 0$*) random variables with* $\mathbf{E}\left[(X_1)^2\right] < \infty$*, and* S_n*,* μ*,* σ *be*

- $S_n := X_1 + \ldots + X_n$,
- $\mu := \mathbf{E}\left[X_1\right]$,
- $\sigma := \sqrt{\mathbf{E}\left[(X_1 - \mu)^2\right]}$,

then $\frac{S_n - n\mu}{\sigma\sqrt{n}}$ *converges in distribution to a unit variance zero mean Gaussian random variable, that is*

$$\mathbf{P}\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq x\right) \to \Phi(x) \qquad \text{as } n \to \infty \qquad \forall x \in \mathbb{R}$$

where $\Phi(x) := \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{\tau^2}{2}} \mathrm{d}\tau$.

Does C.L.T. imply the weak law of large numbers?

$$
\begin{aligned}
\mathbf{P}\left(\left|\frac{S_n-n\mu}{n}\right|>\epsilon\right) &= \mathbf{P}\left(\frac{S_n-n\mu}{\sigma\sqrt{n}}<-\frac{\epsilon\sqrt{n}}{\sigma}\right)+\mathbf{P}\left(\frac{S_n-n\mu}{\sigma\sqrt{n}}>\frac{\epsilon\sqrt{n}}{\sigma}\right)\\
&= \mathbf{P}\left(\frac{S_n-n\mu}{\sigma\sqrt{n}}<-\frac{\epsilon\sqrt{n}}{\sigma}\right)+1-\mathbf{P}\left(\frac{S_n-n\mu}{\sigma\sqrt{n}}\le\frac{\epsilon\sqrt{n}}{\sigma}\right)\\
&\le \left(\Phi\left(-\frac{\epsilon\sqrt{n}}{\sigma}\right)+\Delta(-\frac{\epsilon\sqrt{n}}{\sigma},n)\right)+1-\left(\Phi\left(\frac{\epsilon\sqrt{n}}{\sigma}\right)-\Delta(\frac{\epsilon\sqrt{n}}{\sigma},n)\right)\\
&= \left(\Phi\left(-\frac{\epsilon\sqrt{n}}{\sigma}\right)+\Delta(-\frac{\epsilon\sqrt{n}}{\sigma},n)\right)+\Phi\left(-\frac{\epsilon\sqrt{n}}{\sigma}\right)+\Delta(\frac{\epsilon\sqrt{n}}{\sigma},n)\\
&= 2\Phi\left(-\frac{\epsilon\sqrt{n}}{\sigma}\right)+\Delta(-\frac{\epsilon\sqrt{n}}{\sigma},n)+\Delta(\frac{\epsilon\sqrt{n}}{\sigma},n)
\end{aligned}
$$

where $\Delta(x,n)=\mathbf{P}\left(\frac{S_n-n\mu}{\sqrt{n}}\le x\right)-\Phi(x)$.

$\lim_{n\to\infty}\Delta(x,n)=0\ \forall x$ by C.L.T. and $\lim_{n\to\infty}\Phi(-\epsilon\sqrt{n}/\sigma)=0$.

But "$\lim_{n\to\infty}\Delta(x,n)=0\ \forall x$" $\not\Rightarrow$ "$\lim_{n\to\infty}\Delta(\pm\epsilon\sqrt{n}/\sigma,n)=0$".

Nevertheless $\lim_{n\to\infty}\Delta(\pm\epsilon\sqrt{n}/\sigma,n)=0$ because the convergence in C.L.T. is uniform, i.e. $\sup_{x\in\mathbb{R}}|\Delta(x,n)|\to 0$ as $n\to\infty$.

Why? If $f_n\to f$ pointwise, $f:\mathbb{R}\to\mathbb{R}$ is continuous, and f_n's are monotone, then $f_n\to f$ uniformly.

Proof: **H.W.**

Thus C.L.T. implies

$$\sup_{x\in\mathbb{R}} \left|\mathbf{P}\left(\frac{S_n-n\mu}{\sigma\sqrt{n}} \le x\right) - \Phi(x)\right| \to 0 \qquad \text{as } n \to \infty$$

How fast does $\mathbf{P}\left(\frac{S_n-n\mu}{\sigma\sqrt{n}} \le x\right)$ converge to the Gaussian CDF?

Theorem (Berry Esseen Theorem [Ber41, Ess42, She10])

Let $X_1, X_2, \ldots$ *be independent random variables with finite absolute third moments, i.e.* $\mathbf{E}\left[|X_n|^3\right] < \infty$, *and* $S_n := X_1 + \ldots + X_n$, *then there exists an absolute constant* $C \le 0.5600$ *s.t.*

$$\left|\mathbf{P}\left(\frac{S_n-\mathbf{E}[S_n]}{\sqrt{D_n}} \le x\right) - \Phi(x)\right| \le C\frac{T_n}{(D_n)^{3/2}} \qquad \forall x \in \mathbb{R}$$

where $D_n := \sum_{k=1}^n \mathbf{E}\left[(X_k - \mathbf{E}[X_k])^2\right]$, $T_n := \sum_{k=1}^n \mathbf{E}\left[|X_k - \mathbf{E}[X_k]|^3\right]$, *and* $\Phi(x) := \frac{1}{\sqrt{2\pi}}\int_{-\infty}^x e^{-\tau^2/2}\mathrm{d}\tau$.

For i.i.d. r.v.s

$$\sup_{x\in\mathbb{R}} \left|\mathbf{P}\left(\frac{S_n-n\mu}{\sigma\sqrt{n}} \le x\right) - \Phi(x)\right| \le C\frac{\mathbf{E}\left[|X_1-\mathbf{E}[X_1]|^3\right]}{\sigma^3\sqrt{n}}$$

Theorem (Asymptotic Equipartition Property [CT06, Th.3.1.1])

If $\mathsf{X}_1, \mathsf{X}_2, \ldots$ *are i.i.d.* $\sim p_{\mathsf{X}}$*, then* $\frac{1}{n} \log p_{\mathsf{X}_1^n}(\mathsf{X}_1^n) \xrightarrow{\mathbf{P}} -H(\mathsf{X})$.

Proof.

If $\mathsf{Y}_k := \log p_{\mathsf{X}}(\mathsf{X}_k)$ for all $k \in \mathbb{Z}_+$ then

- $\mathsf{Y}_1, \mathsf{Y}_2, \ldots$ are i.i.d. random variables.
- $\mathbf{E}[\mathsf{Y}_k] = -H(\mathsf{X})$ for all k by definition.
- $\mathbf{E}[|\mathsf{Y}_k|] = H(\mathsf{X})$ for all k because $\mathsf{Y}_k \leq 0$.
- $\frac{1}{n} \sum_{k=1}^n \mathsf{Y}_k \xrightarrow{\mathbf{P}} -H(\mathsf{X})$ by the weak law of large numbers.
- $\sum_{k=1}^n \mathsf{Y}_k = \log p_{\mathsf{X}_1^n}(\mathsf{X}_1^n)$ because $p_{\mathsf{X}_1^n}(\mathsf{X}_1^n) = \prod_{k=1}^n p_{\mathsf{X}}(\mathsf{X}_k)$.

□

We can invoke strong law of large numbers instead: $\frac{1}{n} \log p_{\mathsf{X}_1^n}(\mathsf{X}_1^n) \xrightarrow{a.s.} -H(\mathsf{X})$ under the same hypothesis.

- $\frac{1}{n}\log p_{X_1^n}(X_1^n) \xrightarrow{\mathbf{P}} -H(X)$.
- $\mathbf{P}\left(-H(X)-\epsilon \le \frac{1}{n}\log p_{X_1^n}(X_1^n) \le -H(X)+\epsilon\right) \to 1$ as $n \to \infty$ for all $\epsilon > 0$.
- $\mathbf{P}\left(2^{-n(H(X)+\epsilon)} \le p_{X_1^n}(X_1^n) \le 2^{-n(H(X)-\epsilon)}\right) \to 1$ as $n \to \infty$ for all $\epsilon > 0$.

[CT06] calls this "insidiously self referential, but well defined nonetheless"

Definition (Typical Set)

$$\mathcal{A}_\epsilon^{(n)}(X) := \{x_1^n \in \mathcal{X}_1^n : 2^{-n(H(X)+\epsilon)} \le p_{X_1^n}(x_1^n) \le 2^{-n(H(X)-\epsilon)}\} \tag{4}$$

We use $\mathcal{A}_\epsilon^{(n)}$ instead of $\mathcal{A}_\epsilon^{(n)}(X)$ whenever there is no ambiguity.

Theorem

(a) $\forall \delta > 0$, $\mathbf{P}\left(X_1^n \in \mathcal{A}_\epsilon^{(n)}\right) \ge 1-\delta$ *for sufficiently large n.* *AEP*

(b) $\left|\mathcal{A}_\epsilon^{(n)}\right| \le 2^{n(H(X)+\epsilon)}$. (4) & $\mathbf{P}\left(X_1^n \in \mathcal{A}_\epsilon^{(n)}\right) \le 1$

(c) $\forall \delta > 0$, $\left|\mathcal{A}_\epsilon^{(n)}\right| \ge (1-\delta)2^{n(H(X)-\epsilon)}$ *for sufficiently large n.* (4) & *part* (a)

Corollary

$\left|\frac{1}{n}\log\left|\mathcal{A}_\epsilon^{(n)}\right| - H(X)\right| \le 2\epsilon$ *for all n large enough and* $\left|\frac{1}{n}\log p_{X_1^n}(x_1^n) + H(X)\right| \le \epsilon \;\; \forall x_1^n \in \mathcal{A}_\epsilon^{(n)}$.

Definition

For any $\delta \in (0,1)$ and $n \in \mathbb{Z}_+$ the set $\mathcal{B}_\delta^{(n)} \subset \mathcal{X}_1^n$ is the smallest set, i.e. the one with the fewest number of elements, satisfying $\mathbf{P}\left(\mathsf{X}_1^k \in \mathcal{B}_\delta^{(n)}\right) \geq 1-\delta$.

Theorem

Let $\mathsf{X}_1, \mathsf{X}_2, \ldots$ *be i.i.d.* $\sim p_{\mathsf{X}}$, *then* $\lim_{n\to\infty} \frac{1}{n} \log \left|\mathcal{B}_\delta^{(n)}\right| = H(\mathsf{X})$ *for any* $\delta \in (0,1)$.

Proof.

$\mathbf{P}\left(\mathsf{X}_1^k \in \mathcal{A}_\epsilon^{(n)}\right) \geq 1-\delta$ —hence $\left|\mathcal{B}_\delta^{(n)}\right| \leq \left|\mathcal{A}_\epsilon^{(n)}\right|$— for n large enough by AEP.

On the other hand $\frac{1}{n}\log\left|\mathcal{A}_\epsilon^{(n)}\right| \leq H(\mathsf{X})+\epsilon$. Thus $\frac{1}{n}\log\left|\mathcal{B}_\delta^{(n)}\right| \leq H(\mathsf{X})+\epsilon$ for n large enough.

$$\begin{aligned}
\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{A}_\epsilon^{(n)} \cap \mathcal{B}_\delta^{(n)}\right) &\geq \mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{B}_\delta^{(n)}\right) - \mathbf{P}\left(\mathsf{X}_1^n \notin \mathcal{A}_\epsilon^{(n)}\right) \quad \geq 1-\delta-\mathbf{P}\left(\mathsf{X}_1^n \notin \mathcal{A}_\epsilon^{(n)}\right) \\
&\geq \tfrac{1-\delta}{2} && \text{for n large enough.} \\
\left|\mathcal{A}_\epsilon^{(n)} \cap \mathcal{B}_\delta^{(n)}\right| &\geq \tfrac{1-\delta}{2} 2^{n(H(\mathsf{X})-\epsilon)} && \text{because } p_{\mathsf{X}_1^n}(x_1^n) \leq 2^{-n(H(\mathsf{X})-\epsilon)} \ \forall x_1^n \in \mathcal{A}_\epsilon^{(n)} \\
&\geq 2^{n(H(\mathsf{X})-2\epsilon)} && \text{for n large enough.}
\end{aligned}$$

$\lim_{n\to\infty} \frac{1}{n}\log\left|\mathcal{B}_\delta^{(n)}\right| = H(\mathsf{X})$ because $\forall \epsilon > 0$, $\left|\frac{1}{n}\log\left|\mathcal{B}_\delta^{(n)}\right| - H(\mathsf{X})\right| \leq 2\epsilon$ for large enough n. $\square$

Definition (Jointly Typical Set)

If $(X_1, Y_1), (X_2, Y_2), \ldots$ be i.i.d. $\sim p_{X,Y}$, then (x_1^n, y_1^n) is in the jointly typical set $\mathcal{A}_\epsilon^{(n)}(X, Y)$ iff it satisfies the following three conditions

$$2^{-n(H(X)+\epsilon)} \leq p_{X_1^n}(x_1^n) \leq 2^{-n(H(X)-\epsilon)} \tag{5}$$

$$2^{-n(H(Y)+\epsilon)} \leq p_{Y_1^n}(y_1^n) \leq 2^{-n(H(Y)-\epsilon)} \tag{6}$$

$$2^{-n(H(X,Y)+\epsilon)} \leq p_{X_1^n, Y_1^n}(x_1^n, y_1^n) \leq 2^{-n(H(X,Y)-\epsilon)} \tag{7}$$

Theorem

If $(x_1^n, y_1^n) \in \mathcal{A}_\epsilon^{(n)}(X, Y)$, *then*

(a) $\left| \frac{1}{n} \log p_{X_1^n | Y_1^n}(x_1^n | y_1^n) + H(X|Y) \right| \leq 2\epsilon$ *by (6) and (7)*

(b) $\left| \frac{1}{n} \log \frac{p_{X_1^n, Y_1^n}(x_1^n, y_1^n)}{p_{X_1^n}(x_1^n) p_{Y_1^n}(y_1^n)} - I(X; Y) \right| \leq 3\epsilon$ *by (5) and part (a)*

Theorem

If $(X_1, Y_1), (X_2, Y_2), \ldots$ *be i.i.d.* $\sim p_{X,Y}$*, then for all* $\delta > 0$*,* $\mathbf{P}\left((X_1^n, Y_1^n) \in \mathcal{A}_\epsilon^{(n)}\right) \geq 1 - \delta$ *for* n *large enough.*

Proof.

Let us define set for constraints for $\mathcal{A}_\epsilon^{(n)}$ given in (5), (6), and (7)

$$\mathcal{E}_1 = \{(x_1^n, y_1^n) : 2^{-n(H(X)+\epsilon)} \leq p_{X_1^n}(x_1^n) \leq 2^{-n(H(X)-\epsilon)}\} \qquad \mathcal{E}_1^c = \Omega \setminus \mathcal{E}_1$$

$$\mathcal{E}_2 = \{(x_1^n, y_1^n) : 2^{-n(H(Y)+\epsilon)} \leq p_{Y_1^n}(y_1^n) \leq 2^{-n(H(Y)-\epsilon)}\} \qquad \mathcal{E}_2^c = \Omega \setminus \mathcal{E}_2$$

$$\mathcal{E}_3 = \{(x_1^n, y_1^n) : 2^{-n(H(X,Y)+\epsilon)} \leq p_{X_1^n, Y_1^n}(x_1^n, y_1^n) \leq 2^{-n(H(X,Y)-\epsilon)}\} \qquad \mathcal{E}_3^c = \Omega \setminus \mathcal{E}_3$$

$$\mathcal{A}_\epsilon^{(n)} = \mathcal{E}_1 \cap \mathcal{E}_2 \cap \mathcal{E}_3$$

As a result of weak law of large number for n large enough

$$\mathbf{P}(\mathcal{E}_\imath^c) \leq \delta/3 \qquad \forall \imath \in \{1, 2, 3\}$$

$$\mathbf{P}(\mathcal{E}_1^c \cup \mathcal{E}_2^c \cup \mathcal{E}_3^c) \leq \mathbf{P}(\mathcal{E}_1^c) + \mathbf{P}(\mathcal{E}_2^c) + \mathbf{P}(\mathcal{E}_3^c) \qquad \text{by the union bound}$$

$$\mathbf{P}(\mathcal{E}_1 \cap \mathcal{E}_2 \cap \mathcal{E}_3) = 1 - \mathbf{P}(\mathcal{E}_1^c \cup \mathcal{E}_2^c \cup \mathcal{E}_3^c) \qquad \text{by De Morgan's laws}$$

$$\mathbf{P}(\mathcal{E}_1 \cap \mathcal{E}_2 \cap \mathcal{E}_3) \geq 1 - \delta$$

□

Are there other ways to define typical set? How about Berry-Esseen Theorem?

Drawback: Stronger hypothesis.

Advantage: Sharper bounds.

Theorem

If $X_1, X_2, \ldots$ *are i.i.d.* $\sim p_X$, $\mathbf{E}\left[|\log p_X(X)|^3\right] < \infty$, *and* $\sigma > 0$ *then*

$$\mathbf{P}\left(\left|\log p_{X_1^n}(X_1^n) + nH(X)\right| \leq \tau\sigma\sqrt{n}\right) \geq \Phi(\tau) - \Phi(-\tau) - 2\frac{T}{\sigma^3}\frac{1}{\sqrt{n}} \qquad \forall \tau > 0$$

where $\sigma := \sqrt{\mathbf{E}\left[(\log p_X(X) + H(X))^2\right]}$ *and* $T := \mathbf{E}\left[|\log p_X(X) + H(X)|^3\right]$.

Proof.

Apply Berry-Essen theorem via $Y_k := \log p_X(X_k)$

$$\left|\mathbf{P}\left(\frac{\log p_{X_1^n}(X_1^n) + nH(X)}{\sigma\sqrt{n}} \leq \tau\right) - \Phi(\tau)\right| \leq \frac{T}{\sigma^3}\frac{1}{\sqrt{n}} \qquad \forall \tau \in \mathbb{R}.$$

Use above identity at τ and at $-\tau$

$$\mathbf{P}\left(\frac{\log p_{X_1^n}(X_1^n) + nH(X)}{\sigma\sqrt{n}} \leq \tau\right) \geq \Phi(\tau) - \frac{T}{\sigma^3}\frac{1}{\sqrt{n}} \qquad \forall \tau \in \mathbb{R}$$

$$\mathbf{P}\left(\frac{\log p_{X_1^n}(X_1^n) + nH(X)}{\sigma\sqrt{n}} \leq -\tau\right) \leq \Phi(-\tau) + \frac{T}{\sigma^3}\frac{1}{\sqrt{n}} \qquad \forall \tau \in \mathbb{R}$$

Definition

$$\mathcal{E}_\tau^{(n)} := \left\{ x_1^n \in \mathcal{X}_1^n : 2^{-nH(\mathsf{X})-\sqrt{n}\sigma\tau} \le p_{\mathsf{X}_1^n}(x_1^n) \le 2^{-nH(\mathsf{X})+\sqrt{n}\sigma\tau} \right\}$$

$$\mathcal{E}^{(n)} := \left\{ x_1^n \in \mathcal{X}_1^n : 2^{-nH(\mathsf{X})-\sqrt{n\log n}\sigma} \le p_{\mathsf{X}_1^n}(x_1^n) \le 2^{-nH(\mathsf{X})+\sqrt{n\log n}\sigma} \right\}$$

Theorem

If $\mathsf{X}_1, \mathsf{X}_2, \ldots$ *are i.i.d.* $\sim p_{\mathsf{X}}$ *and* $\mathbf{E}\left[|\log p_{\mathsf{X}}(\mathsf{X})|^3\right] < \infty$ *then*

(a) $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}_\tau^{(n)}\right) \ge \Phi(\tau) - \Phi(-\tau) - 2\frac{T}{\sigma^3}\frac{1}{\sqrt{n}}$.

(b) $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)}\right) \ge 1 - \frac{1}{\sqrt{n}} - 2\frac{T}{\sigma^3}\frac{1}{\sqrt{n}}$.

Proof.

$$\begin{aligned}
\Phi(\tau) - \Phi(-\tau) &= 1 - 2\frac{1}{\sqrt{2\pi}}\int_\tau^\infty e^{-z^2/2}\mathrm{d}z \\
&= 1 - 2\frac{e^{-\tau^2/2}}{\sqrt{2\pi}}\int_0^\infty e^{-s^2/2}e^{-s\tau}\mathrm{d}s \qquad \text{(by } z = s+\tau\text{)} \\
&\ge 1 - e^{-\tau^2/2}
\end{aligned}$$

$$\Phi\left(\sqrt{\log n}\right) - \Phi\left(-\sqrt{\log n}\right) \ge 1 - \frac{1}{\sqrt{n}}$$

Theorem

If $X_1, X_2, \ldots$ *are i.i.d.* $\sim p_X$ *and* $\mathbf{E}\left[|\log p_X(X)|^3\right] < \infty$ *then*

$$\mathbf{P}\left(X_1^n \in \mathcal{E}^{(n)}\right) \geq 1 - \left(1 + 2\frac{T}{\sigma^3}\right)\frac{1}{\sqrt{n}}$$

where $\mathcal{E}^{(n)} := \left\{ x_1^n \in \mathcal{X}_1^n : 2^{-n\left(H(X)+\sigma\sqrt{\frac{\log n}{n}}\right)} \leq p_{X_1^n}(x_1^n) \leq 2^{-n\left(H(X)-\sigma\sqrt{\frac{\log n}{n}}\right)} \right\}$,

$\sigma := \sqrt{\mathbf{E}\left[(\log p_X(X) + H(X))^2\right]}$, *and* $T := \mathbf{E}\left[|\log p_X(X) + H(X)|^3\right]$.

Why $\sigma = 0$ case is included?

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